MERCK MANUAL
Professional Version

Nonthrombotic Pulmonary Embolism

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Nonthrombotic sources of pulmonary embolism include air, fat, amniotic fluid, infected material, foreign bodies, tumors, and orthopedic cement.

Pulmonary embolism (PE) can arise from nonthrombotic sources. PE caused by nonthrombotic sources results in clinical syndromes that differ from those caused by thrombotic PE. Diagnosis is usually based in part or completely on clinical criteria, including, in particular, the patient's risk. Treatment includes supportive measures.

Air embolism

Air embolism is caused by introduction of large amounts of air into systemic veins or into the right side of the heart, which then move to the pulmonary arterial system. Pulmonary outflow tract obstruction may occur, which can be rapidly fatal.

Causes include

- Surgery
- Blunt trauma
- Defective or uncapped venous catheters
- Errors occurring during insertion or removal of central venous catheters

Treatment includes placement of the patient in the left lateral decubitus position, preferably in the Trendelenburg position (ie, head lower than the feet), to trap air in the apex of the right ventricle and thus prevent brain embolism if there is a right-to-left shunt or right ventricular and pulmonary artery outflow obstruction. Supportive measures are also needed.

Rapid decompression after underwater diving may cause microbubble formation in the pulmonary circulation, a different problem, which results in endothelial damage, hypoxemia, and diffuse infiltrates (see Arterial Gas Embolism).

Fat embolism

Fat embolism is caused by the introduction of fat or bone marrow particles into the systemic venous system and then into pulmonary arteries. Causes include fractures of long bones, orthopedic procedures, microvascular occlusion or necrosis of bone marrow in patients with sickle cell crisis, and, rarely, toxic modification of native or parenteral serum lipids. Certain procedures such as liposuction, particularly with fat grafting, place patients at risk.

Fat embolism causes a pulmonary syndrome similar to acute respiratory distress syndrome (ARDS), with severe hypoxemia of rapid onset often accompanied by neurologic changes and a petechial rash.

Early splinting of fractures of long bones and operative rather than external fixation are thought to help prevent fat embolism.

Amniotic fluid embolism

Amniotic fluid embolism is a rare syndrome caused by introduction of amniotic fluid into the maternal venous and then pulmonary arterial system. The syndrome occurs around the time of labor (amniotic fluid embolism) or, even less often, during prepartum uterine manipulations.

Patients can have cardiac and respiratory distress due to anaphylaxis, vasoconstriction causing acute severe pulmonary hypertension, and direct pulmonary microvascular toxicity with hypoxemia and pulmonary infiltrates.

Septic embolism

Septic embolism occurs when infected material embolizes to the lung. Causes include IV drug use, right-sided infective endocarditis, and septic thrombophlebitis.

Septic embolism causes symptoms and signs of pneumonia (eg, fever, cough, sputum production, pleuritic chest pain, dyspnea, tachypnea, and tachycardia) or sepsis (eg, fever, hypotension, oliguria, tachypnea, tachycardia, and confusion). Initially, nodular opacities appear on the chest radiograph; the appearance may progress to peripheral infiltrates, and emboli may cavitate (particularly emboli caused by *Staphylococcus aureus*).

Treatment includes that of the underlying infection.

Foreign body embolism

Foreign body embolism caused by introduction of particulate matter into the pulmonary arterial system, usually by IV injection of inorganic substances, such as talc by heroin users or elemental mercury by patients with psychiatric disorders.

Focal pulmonary infiltrates may result.

Tumor embolism

Tumor embolism is a rare complication of cancer (usually adenocarcinoma) in which neoplastic cells from an organ enter the systemic venous circulation and pulmonary arterial system where they lodge, proliferate, and obstruct blood flow. Benign metastasizing leiomyoma may also embolize to the lungs.

Patients typically present with dyspnea and pleuritic chest pain and signs of cor pulmonale that develop over weeks to months.

The diagnosis may be suggested by micronodules or diffuse pulmonary infiltrates on chest radiograph or CT scan, but these findings are neither very sensitive nor specific. The diagnosis can be confirmed by biopsy or occasionally by cytologic aspiration and histologic study of pulmonary capillary blood.

Cement embolism

Cement embolism may develop after certain procedures such as vertebroplasty.